

M GREGOR M N D E L

The Augustinian monk Gregor Mendel is widely regarded as the founder of modern genetics. He discovered the laws of heredity and realized that family traits are inherited through units that are now known as genes.

Johann Mendel was born to peasant farmers in a village in what is now the Czech Republic. He initially trained in mathematics and philosophy before entering the priesthood as a way of furthering his education. Renamed Gregor when he joined St. Thomas's Abbey in Brno, Mendel was sent to Vienna University to complete his studies, after which he returned to teach at the abbey.

Mendel was interested in plant breeding and heredity. Inspired to help the monks improve their crops, he focused on peas. His mathematical background helped him to design his experiments: he looked at "either/or" traits, such as seed color, flower color, and plant height, which made it easier for him to interpret his results.

Generations of pea plants

In one example, Mendel crossed green-seeded and yellow-seeded peas, and all the seeds in the next generation were yellow. He then crossed two of their offspring with each other and found that the

“We have now the means of beginning **an analysis of living organisms.**”

William Bateson, 1913

MILESTONES

FURTHER EDUCATION

Works as a teacher while a friar, then goes to study at the University of Vienna in 1851.

FIRST EXPERIMENTS

Authorized to study the St. Thomas's Abbey garden in 1854; begins to cultivate peas in 1856.

SEMINAL PAPER

Publishes his *Experiments in Plant Hybridization* in 1866, but its significance is unappreciated.

HIGHER CALLING

Becomes Abbot of St. Thomas's in 1868, and puts aside his scientific research.